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AI & DATA SCIENCE DIPLOMA · MACHINE LEARNING MODULE

Machine Learning Capstone Project

Telecom Customer Intelligence

Supervised Learning + Unsupervised Learning — one project, one story

Student Brief · 3 CSV files · 8 parts · 100 points

START HERE

The story

You have just joined the data team at **Nile Connect**, a mobile and home-internet operator in Egypt. The CEO has three worries and one question for you:

- **Too many customers are leaving.** About 1 in 4 customers in our sample has left. Who is next?
- **Billing is a black box.** Why does one customer pay 200 EGP and another 600 EGP?
- **One offer for everyone does not work.** We send the same SMS to all customers. Who are our customers, really?
- **Our bundles are random.** Which services do people buy together?

Your job: use everything you learned in the Machine Learning module to answer these four questions with data, and write a short, clear recommendation the CEO can act on.

> **Think like a data scientist, not like a student**
> A model with a high score is not the goal. A decision the business can trust is the goal. In every part, ask yourself: *"So what? What would we do differently on Monday morning?"*

What you will practice

Topic from the course	Where you use it
Linear Regression (one + multiple variables)	Part 2 - predict the monthly bill
R2, RMSE, MAE	Part 2 - judge the bill models
Logistic Regression, Decision Tree, SVM, KNN, Naive Bayes	Part 3 - predict churn
Ensemble Learning: Random Forest, XGBoost	Part 3 + 4 - stronger churn models
Accuracy, Precision, Recall, F1-Score	Part 3 - judge churn models, pick the right metric
Cross Validation, GridSearchCV, RandomizedSearchCV	Part 4 - fair comparison and tuning
Clustering: KMeans, HAC, DBSCAN	Part 5 - find customer segments
Dimensionality Reduction: PCA	Part 5 - see and simplify the segments
Association Rules: Apriori	Part 6 - bundle recommendations

Time: about 10-14 hours of work. Work alone or in pairs. Tools: Python, pandas, scikit-learn, xgboost, mlxtend, matplotlib / seaborn.

THE DATA

Your three CSV files

All three files describe the same company. They are **not clean** - just like real data. Do not trust a column until you have looked at it.

1) customers.csv - 3,015 rows, 18 columns (used in Parts 1-4 and 7)

Column	Meaning
customer_id	Unique ID (C00001...). An ID, not a feature.
age, gender, governorate	Who the customer is and where they live.
tenure_months	How many months the customer has been with us.
contract_type, plan, payment_method, device_type	How the customer buys and pays.
has_home_internet	Yes / No - also has our home internet.
data_usage_gb, voice_minutes, sms_count	Monthly usage.
num_services, support_calls, late_payments	Number of services, calls to support, late payments.
monthly_bill_egp	**TARGET for Part 2 (regression).** Monthly bill in EGP.
churn	**TARGET for Parts 3-4 (classification).** Yes = customer left.

2) usage_behavior.csv - 2,000 rows, 13 columns (used in Part 5 and Part 7)

Habits of the first 2,000 customers (same customer_id as above). **There is no label.** You must find the groups yourself.

Column	Meaning
avg_daily_data_mb, night_usage_pct	Daily data use, and the share used at night.
streaming_hours_week, social_media_hours_week, gaming_hours_week, video_call_hours_week	Weekly hours by app type.
calls_per_day, avg_call_minutes	Voice habits.
intl_calls_month, roaming_days_year	International calls and travel.
topups_per_month, app_count	Top-ups and number of apps used.

3) service_transactions.csv - 12,993 rows, 2 columns (used in Part 6)

One row = one service in one basket. Columns: `transaction_id` (4,500 baskets) and `service` (13 services, e.g. Home_Internet, Router_Upgrade, Roaming_Pack...). You must reshape it into a basket table before Apriori.

> **A warning from experience**
> Spend your first hour only on looking at the data. Most students jump to models in 10 minutes, then wonder why the results look strange. The dirt is there on purpose: there are at least **six** different problems hidden across the files. Part 1 asks you to find and list them.

PART 1 · 10 POINTS

Know your data and clean it

Files: customers.csv, usage_behavior.csv, service_transactions.csv

1.1	Load the three files. For each one print shape, data types, missing values per column and basic statistics. Write 3-5 lines on what you see.
1.2	**Data audit.** Make a small table: *problem found* | *which file / column* | *how many rows* | *how you fixed it*. Look for missing values, duplicate rows, impossible or extreme values, and text that is written in different ways. *Hint: Check value_counts() of every text column. Look at the max of every number column.*
1.3	Clean the data. Choose a different rule for each type of problem (drop, median, mode, cap, fix text) and **explain why** in one sentence each.
1.4	Explore. Draw at least 5 charts. Required: churn rate by contract_type, churn rate by tenure group, and a correlation heatmap of the numeric columns. Write one business sentence under each chart.
1.5	Look at the churn column. What percent of customers left? Why can this make **accuracy** a dangerous metric? Answer in 3 lines.

Rule
Every cleaning choice must be written down. Data leakage rule: fit anything that learns from data (median, scaler, encoder) on the **training set only**. For Part 1, simple cleaning before the split is fine; for Parts 3-4, use a Pipeline.

PART 2 · 10 POINTS

Predict the monthly bill (Regression)

Target: monthly_bill_egp · File: customers.csv

2.1	**One variable.** Fit a Linear Regression with only `data_usage_gb`. Plot the points and the line. In plain words: what does the slope mean in EGP?
2.2	**Multiple variables.** Add more columns (usage, num_services, plan, has_home_internet...). Encode the text columns. Split 80/20 with random_state=42.
2.3	Report **R2, RMSE and MAE** on the test set for both models. Put them in one table.
2.4	Explain the three metrics to a non-technical manager in one sentence each. When would MAE be better than RMSE for this company?
2.5	Look at the coefficients. Which 3 features move the bill the most? Does the sign make business sense? Plot residuals - do you see a pattern?

PART 3 · 20 POINTS

Who will leave? (Classification)

Target: churn · File: customers.csv

3.1	Prepare the data: drop customer_id and the target from X, turn churn into 0/1, encode categories, split 80/20 with **stratify=y** and random_state=42. Keep the test set untouched until Part 4 ends.
3.2	Train these **7 models**: Logistic Regression, Decision Tree, SVM, KNN, Naive Bayes, Random Forest, XGBoost. Use default settings first. *Hint: SVM, KNN and Logistic Regression need scaled data. Trees and XGBoost do not. Use a Pipeline so scaling is learned from training data only.*
3.3	For each model report **Accuracy, Precision, Recall and F1** for the churn class. Show them in one table, sorted by F1.
3.4	Draw the confusion matrix of the best 2 models. Explain in your own words: what is a False Negative here, and what is a False Positive? Which one costs the company more?
3.5	**Business cost.** Assume: losing a customer costs 1,500 EGP; a retention offer to one customer costs 200 EGP. Using confusion matrices, calculate the money impact of the 3 best models. Is the best-F1 model also the best-money model?
3.6	Show feature importance from Random Forest **and** XGBoost. Do they agree? Write the top 5 reasons customers leave, in simple language.

Lived-experience tip
A model that says "nobody will churn" gets about 74% accuracy on this data - and is completely useless. Always compare against this dumb baseline before you celebrate any score.

PART 4 · 15 POINTS

Fair comparison and tuning

Cross Validation · GridSearchCV · RandomizedSearchCV

4.1	Use **5-fold Stratified Cross Validation** on the training set for all 7 models. Report mean and standard deviation of F1. Which model is the most *stable*, not only the highest?
4.2	**GridSearchCV** on one model of your choice (Decision Tree or SVM are good). Define a grid of at least 3 parameters, score with F1, cv=5. Show the best parameters and the best score.
4.3	**RandomizedSearchCV** on Random Forest or XGBoost with at least 5 parameters and n_iter of 30 or more. Measure the time it took.
4.4	Compare Grid and Randomized search: time, score, number of combinations tried. When would you pick each one in a real project?
4.5	Choose **one final model**. Evaluate it on the untouched test set **one time only**. Report all four metrics and compare with the cross-validation score. Are they close? If not, what could be wrong?
4.6	Bonus idea (not required): try class_weight='balanced' or a different decision threshold (for example 0.35 instead of 0.5). Does recall improve? What do you lose?

Common mistake
Tuning on the test set. If you check the test score, change something, and check again, the test set is no longer a test. It becomes part of training and your final number is too optimistic.

PART 5 · 20 POINTS

Find the hidden customer segments (Clustering + PCA)

File: usage_behavior.csv · no labels

5.1	Prepare: drop customer_id, handle missing values, and **scale all features** with StandardScaler. Write one sentence: why must we scale before KMeans, HAC and DBSCAN?
5.2	**KMeans.** Try k from 2 to 8. Plot the elbow curve and the silhouette score. Choose k and defend your choice. Then profile each cluster (mean of each feature).
5.3	**HAC.** Draw a dendrogram (use Ward linkage first). Cut it into the same number of clusters. Try at least one other linkage (average or complete). Do the groups match KMeans?
5.4	**DBSCAN.** Choose eps with a k-distance plot and justify min_samples. How many clusters and how many noise points do you get? Look at the noise points: who are these customers? *Hint: If you get one giant cluster or 100% noise, your eps is wrong. Try again with the plot.*
5.5	**PCA.** Fit PCA on the scaled data. Plot explained variance (cumulative). How many components keep about 90% of the information? Draw a 2D scatter plot colored by your best clustering.
5.6	Run KMeans again on the PCA output (the components you kept). Do you get the same segments? What did you gain and what did you lose?
5.7	**Name your segments.** Give each cluster a business name (for example "Night Gamers") and write 2 lines on what offer fits each one.

PART 6 · 10 POINTS

Which services go together? (Association Rules)

File: service_transactions.csv

6.1	Reshape the file into a basket table: one row per transaction, one True/False column per service. *Hint: pd.crosstab or groupby + unstack. mlxtend wants True/False values, not 0/1 with a wrong type.*
6.2	Which 5 services are the most popular? Plot their support.
6.3	Run **Apriori** with min_support around 0.03. Then build rules with min_confidence 0.5. Show the table sorted by **lift**.
6.4	Explain Support, Confidence and Lift with **one real rule from your table**. Why is a rule with high confidence but lift near 1 not interesting?
6.5	Pick the 3 best rules. For each one, write a concrete offer (for example: "Customers who buy X get a discount on Y") and who it should be sent to.

PART 7 · 5 POINTS

Put the two worlds together

Supervised + Unsupervised

7.1	Add the segment from Part 5 to customers.csv (use customer_id; only the first 2,000 customers have one). Show the **churn rate per segment**. Which segment is most at risk?
7.2	On those 2,000 customers, add the segment as a feature to your churn model. Does the score improve? Write an honest answer. A "no" is a fine answer if you can explain it.

PART 8 · 10 POINTS

Tell the story (Final report)

Written for the CEO, not for a data scientist

8.1	Write a **one-to-two page** summary with four headings that match the CEO's four worries: *Churn*, *Billing*, *Segments*, *Bundles*. Each section: the finding, the number that proves it, and the action you recommend.
8.2	Add a short "Limits and next steps" box: what is missing in this data, and what would you collect next?

What to submit

- **One Jupyter notebook** (.ipynb) that runs from top to bottom without errors. Use markdown cells with headings Part 1 … Part 8.
- **The report** as PDF or a short slide deck (2 pages or 5 slides).
- Use `random_state=42` everywhere so your results can be repeated.
- Name the files: `Name_ML_Capstone.ipynb` and `Name_ML_Capstone_Report.pdf`.

Grading

Part	What we look for	Points
1 Data audit	Found the real problems, sensible fixes, clear reasons	10
2 Regression	Correct models and metrics, honest reading of coefficients	10
3 Classification	7 models, right metrics, business cost thinking	20
4 CV and tuning	No leakage, fair comparison, test set used once	15
5 Clustering + PCA	Scaling, justified k / eps, named segments	20
6 Apriori	Correct basket table, rules read with lift, real offers	10
7 Combine	Segment-level churn, honest result	5
8 Report + code	Clear story for the CEO, clean and readable notebook	10
Total		**100**

Before you submit - checklist

- Did you compare every model with the "nobody churns" baseline?
- Did you fit scalers, encoders and imputers only on training data?
- Did you scale the data before KMeans, HAC and DBSCAN?
- Did you remove customer_id from the features?
- Did you touch the test set only once?
- Does every chart have a title, labeled axes and one sentence of meaning?
- Does your notebook run from top to bottom after "Restart and Run All"?

> **Last word**
> There is no single perfect answer here. Two students can choose different models and different k and both get full marks, if they can explain their choices with evidence. Show your thinking.